

1 Components

Figure 1 shows the model, Figure 2 the field separation, Figure 3 the development ladder, and Figure 4 the same-data comparison. Tables 1, 2, 4, 3, and 5 give the numbers.

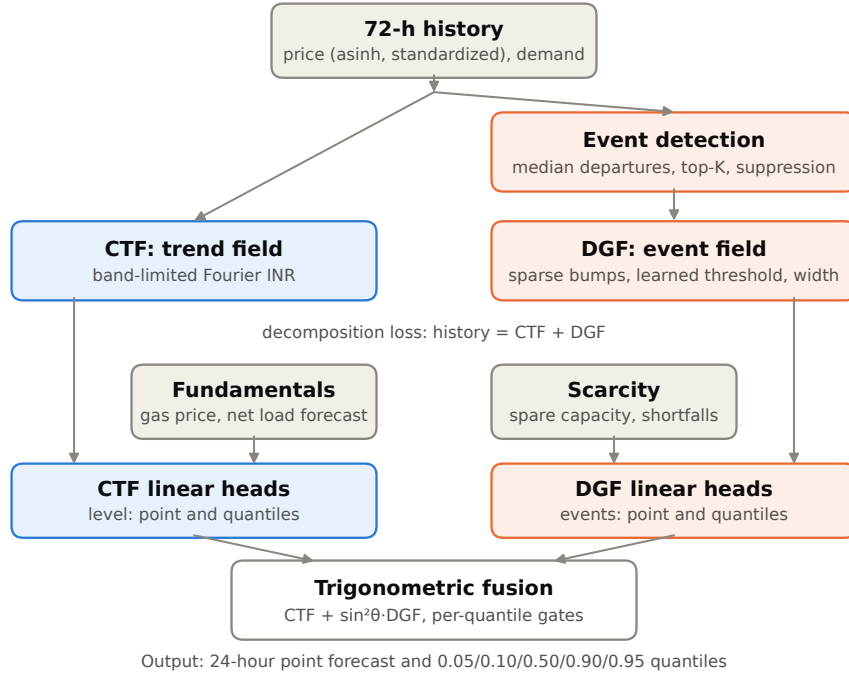


Figure 1: Detected-event dual-field forecaster. Events detected in the 72-hour history, as the largest departures from the window median after non-maximum suppression, form the discrete event field (DGF); the continuous trend field (CTF) fits the remainder under the decomposition loss. Field-specific linear heads read field-matched AEMO forecasts (gas price and net load for the CTF, spare capacity and shortfalls for the DGF) and are fused by a trigonometric gate with per-quantile variants.

Table 1: Same-data comparison on the 2023–2024 test set (17,521 hourly origins per region, 24-hour horizon). All models use the same splits and inputs, without news; cells are means over three seeds, and Mean MAE shows the seed standard deviation. Window RMSE is averaged over forecast windows; 90% AIS is the average interval score of the 5–95% interval; CRPS is approximated by twice the mean pinball loss over the quantiles 0.05, 0.10, 0.50, 0.90, 0.95. Lower is better; best in bold.

Model	MAE			Mean MAE	Window RMSE	90% AIS	CRPS
	NSW	QLD	TAS				
<i>Raw prices</i>							
Ours	48.31	42.53	33.79	41.54 $\pm$ 0.06	78.80	403.5	28.15
XGBoost	50.85	44.70	33.89	43.15 $\pm$ 0.07	81.65	414.3	27.69
GRU	52.35	46.21	38.39	45.65 $\pm$ 1.16	83.43	452.5	30.12
DeepAR	54.24	50.84	38.51	47.86 $\pm$ 1.69	86.09	444.8	30.17
<i>Capped at 650</i>							
Ours	29.79	30.38	30.80	30.32 $\pm$ 0.12	41.29	213.2	17.86
XGBoost	32.88	31.18	30.77	31.61 $\pm$ 0.02	43.37	201.9	16.74
GRU	34.25	34.06	35.54	34.62 $\pm$ 0.62	46.09	276.5	20.39
DeepAR	37.09	38.75	35.19	37.01 $\pm$ 3.02	49.03	241.9	19.64

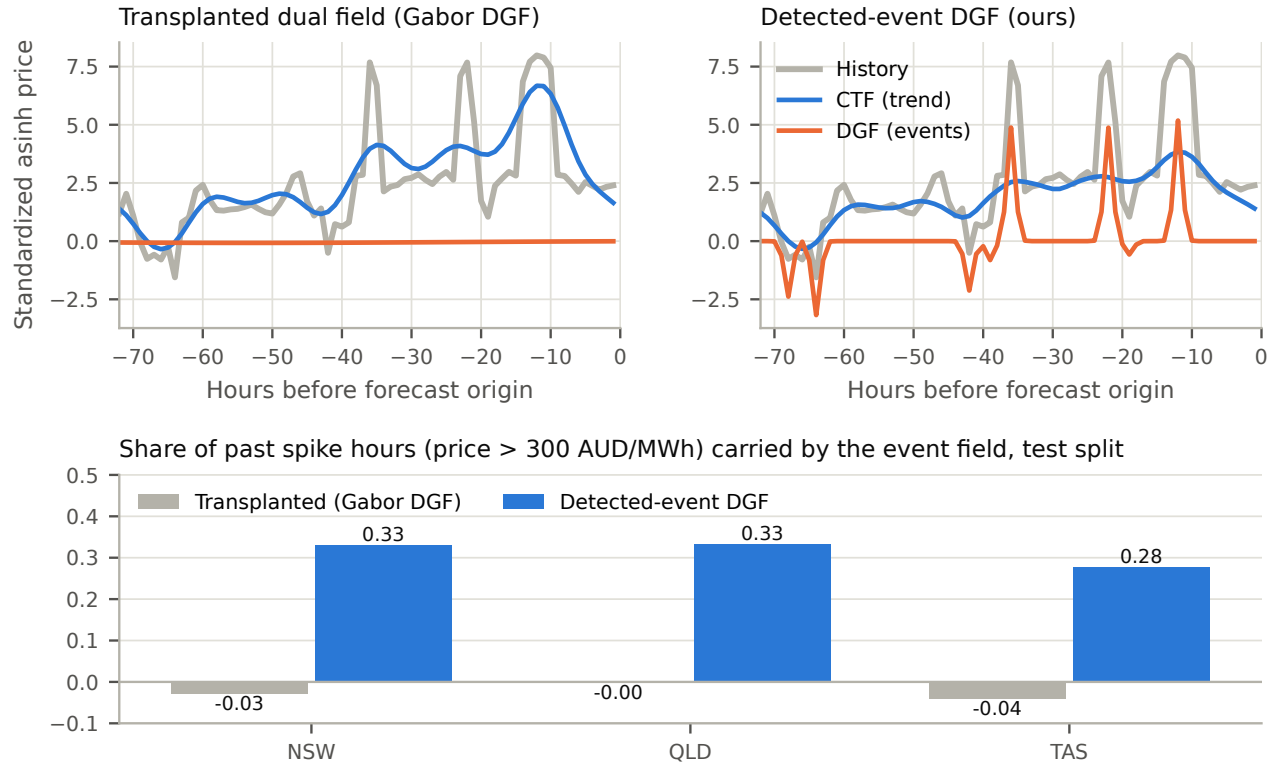


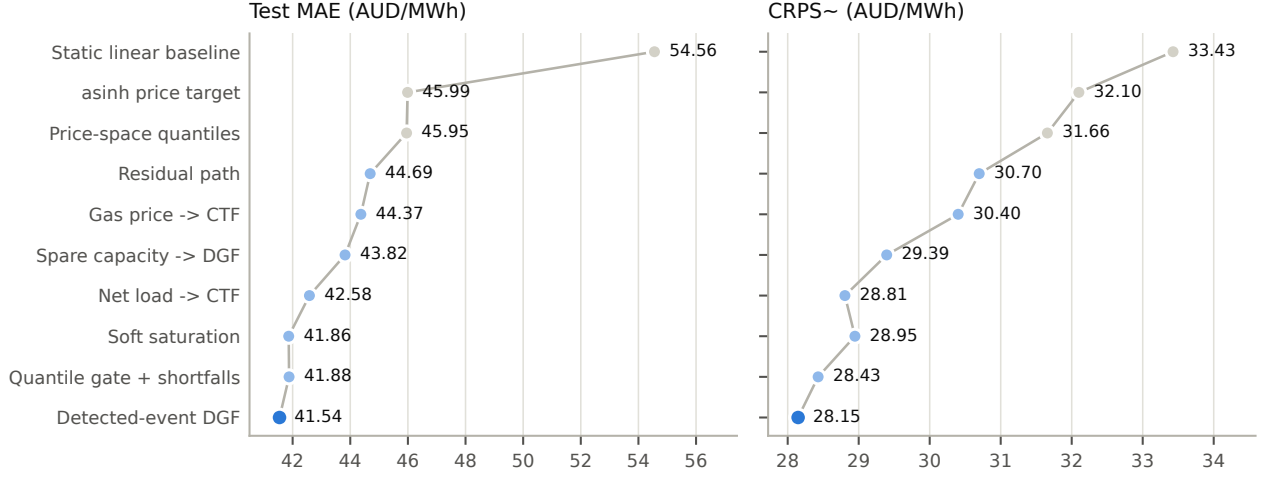
Figure 2: Field separation. *Top*: one NSW test window in standardized asinh price. Transplanted from reconstruction, the Gabor DGF stays flat and the CTF absorbs the spikes (left); with detected events the spikes and troughs move into the DGF and the CTF keeps the level (right). *Bottom*: share of past spike-hour prices (> 300 AUD/MWh) carried by the event field on the test split: -0.04 to 0.00 for the transplanted model, 0.28 to 0.33 for ours.

Table 2: Diebold–Mariano tests of absolute error, ours minus comparator (negative favours ours). Pooled uses the three-region mean loss; DM is its test statistic. $*p < 0.05$, $**p < 0.01$.

Comparator	Pooled Δ MAE	DM	NSW	QLD	TAS
<i>Raw prices</i>					
Transplanted dual field	-0.33**	-4.75	-0.24*	-0.03	-0.72**
No decomposition	-0.35**	-4.39	-0.31*	+0.18	-0.93**
XGBoost	-1.60**	-6.58	-2.54**	-2.17**	-0.09
GRU	-4.11**	-15.50	-4.05**	-3.68**	-4.60**
DeepAR	-6.32**	-16.99	-5.93**	-8.30**	-4.71**
GBDT (same inputs)	-0.57**	-3.09	-1.63**	-0.05	-0.04
Seasonal naive	-15.63**	-6.47	-24.21**	-16.63**	-6.04**
<i>Capped at 650</i>					
Previous trunk	-0.57**	-8.77	-0.26**	-0.32**	-1.14**
XGBoost	-1.29**	-6.88	-3.09**	-0.81**	+0.03
GRU	-4.30**	-18.23	-4.47**	-3.68**	-4.74**
DeepAR	-6.69**	-18.76	-7.30**	-8.38**	-4.39**
GBDT (same inputs)	-0.91**	-5.34	-2.22**	-0.42*	-0.08
Seasonal naive	-6.78**	-16.21	-9.84**	-7.38**	-3.12**

Table 3: Share of past spike-hour prices (> 300 AUD/MWh) carried by the event field on the test split, transplanted Gabor DGF against the detected-event DGF.

Event field	NSW	QLD	TAS
Transplanted (Gabor)	-0.03	-0.00	-0.04
Detected events (ours)	0.33	0.33	0.28



Points: grey = seed 2026 only; light blue = mean of seeds 2026-2028; blue = final model (three seeds). Axes do not start at zero.

Figure 3: Development ladder: three-region mean test MAE and CRPS on raw prices. Each step changes one component of the step above. Grey points are seed 2026 only; blue points are means over seeds 2026-2028. Axes do not start at zero.

Table 4: Ablations on raw prices, three-region means over seeds 2026-2028 (MAE $\pm$ seed standard deviation). The last column gives the paired MAE difference to ours for each seed; positive means the variant is worse.

Variant	MAE	Window RMSE	CRPS	Δ MAE vs ours (seeds)
<i>Structure</i>				
Transplanted dual field (Gabor DGF)	41.88 $\pm$ 0.22	79.21	28.43	+0.38 / +0.03 / +0.58
No decomposition (heads read raw history)	41.90 $\pm$ 0.11	79.31	28.40	+0.30 / +0.25 / +0.51
Adaptive attention atoms + echo	41.84 $\pm$ 0.05	79.01	28.09	+0.30 / +0.28 / +0.30
Detected events + daily echo	42.21 $\pm$ 0.11	79.93	28.22	+0.79 / +0.47 / +0.73
Detected-event DGF (ours)	41.54 $\pm$ 0.06	78.80	28.15	–
<i>Detector settings (ours: 8 events, 3-hour separation, learned threshold, median)</i>				
4 events	41.65 $\pm$ 0.03	78.85	28.25	+0.13 / +0.01 / +0.18
16 events	41.50 $\pm$ 0.08	78.76	28.15	-0.08 / -0.03 / -0.03
No suppression (1-hour separation)	41.85 $\pm$ 0.03	79.00	28.05	+0.36 / +0.21 / +0.35
6-hour separation	41.51 $\pm$ 0.05	78.77	28.28	-0.05 / -0.11 / +0.05
No threshold	41.63 $\pm$ 0.08	78.98	27.97	+0.21 / -0.03 / +0.07
Mean baseline	41.64 $\pm$ 0.03	78.91	28.02	+0.09 / +0.05 / +0.15

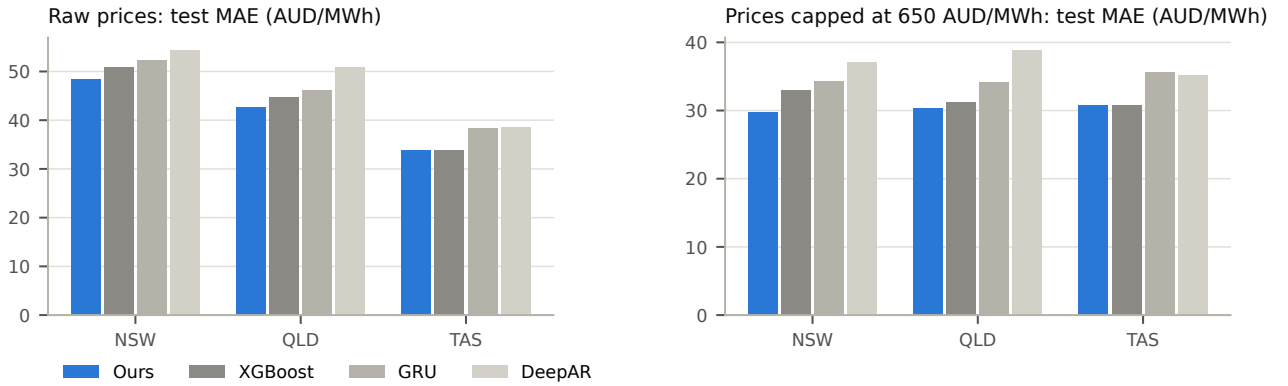


Figure 4: Test MAE per region against re-implemented XGBoost, GRU, and DeepAR on raw prices (left) and prices capped at 650 AUD/MWh (right). All models share the splits, inputs, and scoring; bars are means over three seeds.

Table 5: Ours on prices capped at 650 AUD/MWh beside the numbers reported by RE-Price (its Tables 2–3). The reported models use self-collected news and an unstated spike treatment; the 650 cap is inferred from their error ratios. This is a reference, not a same-data comparison.

Model	MAE				CRPS			
	NSW	QLD	TAS	Mean	NSW	QLD	TAS	Mean
Ours (no news, capped at 650)	29.79	30.38	30.80	30.32	17.39	18.52	17.68	17.86
RE-Price (reported, with news)	23.48	25.85	18.96	22.76	17.36	20.58	13.13	17.02
GPT4TS (reported, with news)	28.12	29.80	22.88	26.93	21.70	24.51	16.75	20.99
Informer (reported, with news)	29.25	31.80	30.85	30.63	22.68	21.78	20.55	21.67
DeepAR (reported, with news)	33.77	36.67	26.51	32.32	26.25	25.94	20.18	24.12
GRU (reported, with news)	33.24	43.24	30.40	35.63	25.12	28.48	19.66	24.42
XGBoost (reported, with news)	38.63	44.70	37.56	40.30	28.79	29.17	23.43	27.13